

1. Set up your event

Before you begin

Use the current Ref OS website. Have the event code, event name, dates, competition program, team roster and volunteer roles available. One authorized person should handle initial setup. The TM connector also needs one Windows or Android device with access to the TM network and the internet.

Create the event

1. On Choose VEX Event, choose Create VEX Event.
2. Enter the full RE or VE event code and use Look up. Alternatively, search USA events by state and date. Review the returned event before creating it.
3. Choose V5RC / VEX U or VEX IQ. Choose Tournament for a single tournament, or League for a series of sessions.
4. Set the event Admin access code, confirm the details and create the event. If lookup is unavailable, create it manually and import the roster later.

Configure the workspace

Open Event Command Center. Use Event Settings to confirm the name, short name, branding, field names and robot-photo requirements. Set the event timezone where available so schedules use the intended local time. For a League, create its sessions and select or start the intended session before importing data. Matches with the same number in different sessions are separate records.

Give volunteers access

Configure Volunteer Access Codes for the event roles. Share each code only with the appropriate crew. Referees use match and violation tools; Inspection uses Robots and Rules; Judge Advisors use Judging and alliance information; Emcees have viewing access. Admins and the Developer manage setup and sync. Ask volunteers to confirm their name and assignment.

Import and verify

Open TM Sync Center in Event Command Center. Import a TM event package or individual exports for teams, schedule, rankings, Skills, alliances and scores. Review the changes before applying them. For IQ, use the IQ schedule/results import in Matches; the live V5 TM API workflow in this guide does not apply to IQ. Open the event on a second device and confirm the roster and schedule agree.

2. Use the VEX Events API

What it does

The VEX Events API reads published event data over the internet. It can find events and teams, import qualification rankings and Skills, and update scores for matches already in Ref OS. Publication can lag the venue. It does not identify the match currently running on a local field; use the TM connector for that.

Start or manage Sync everything

1. Sign in as an Admin or Developer and open the event.
2. In Event Command Center, open the VEX API / Sync everything setup. Its launcher is kept out of the main working tabs.
3. Confirm the full VEX event code, then choose Load divisions.
4. Choose the desired division, or All divisions, one after another for a multi-division event.
5. Choose Start all syncs. Rankings, Skills and existing-match scores run as separate categories. Skills covers the whole event.
6. Use the category Manage/Stop controls in the Command Center when you need to inspect or stop an update.

Automatic startup

With a valid saved event code, the current app attempts to start VEX sync automatically when an Admin or Developer opens the event. This is independent of whether the TM connector is enabled. If automatic startup cannot discover the event or divisions, open the setup, correct the code and start it manually.

Refresh and source behavior

VEX checks run every 15 seconds while the app is open, visible and online. They do not run in a closed or suspended browser. Category errors can leave other categories running. VEX score sync updates existing matches; it does not create the missing schedule. Empty responses retain imported data. TM can provide newer local results before they are published by VEX.

Verify the result

Check Rankings, Skills Challenge and several completed matches. If the values differ from TM, check that the event code, division and League session are correct. Allow for publication delay. If rankings remain stale despite fresh TM scores, inspect the TM rankings in TM itself and the relevant sync status. Do not treat an old published snapshot as a new referee ruling.

Credential handling

Event staff do not enter developer client IDs or client secrets. Those credentials belong in the server-side configuration. Never put them in this guide, screenshots, shared event settings or volunteer messages.

3. Connect Tournament Manager

Choose one connector device

Use one event Admin or Developer device to send TM data to Ref OS Cloud. The device must reach the TM server and the internet. Other volunteers can use the normal Ref OS website or PWA from anywhere; they do not need the TM API key or TM network access. Being on the venue Wi-Fi alone does not make a browser-only PWA a native TM connector.

Prepare TM

In Tournament Manager, open Tools, Options, then Web Publishing. Enable the Local TM API and copy the event API key. Confirm the TM server address and port with the event operator. A plain IP is accepted; include the port if the server uses a port other than the default. Do not expose the TM server to the public internet to make remote Ref OS viewing work.

Windows setup

1. Download the current Ref OS TM Connect Windows executable from the Ref OS connection screen or release download.
2. Double-click the executable. Enter the HTTPS address of the Ref OS website you normally use.
3. Sign in, select the event and confirm the League session, if applicable.
4. Open the TM connection setup, enter the TM address and event API key, then Connect to TM.
5. Select the TM division, Review TM data and choose a TM session if asked. Check the event and preview before Start TM syncing.

Android setup

Download and install the current Ref OS TM Connect APK. Open it, sign in and select the event. Enter the TM address and event API key, review the division and start sync. Keep the app open and the device awake. A newly rebuilt APK must be installed to receive changes to the bundled Android app; refreshing the website cannot update it.

Sync behavior

Qualification data refreshes every 30 seconds. A division with an elimination schedule refreshes every 15 seconds. A live match start schedules an additional score refresh after 30 seconds to collect previous-match results. Missing TM matches are added; existing team assignments and field labels are preserved. Rankings and completed scores come from TM.

Keep it running

Windows continues while minimized; closing its window during sync hides it in the tray. Reopen it from the tray, or use Quit and stop syncing to exit. Android needs to remain open and awake. Leaving the event, signing out, reloading or explicitly stopping sync ends the connection. Event keys stay in memory and must be entered again after a reload.

4. Work with multiple divisions

One server, two divisions

The connector supports cycling two TM divisions from one server. Enter one IP address and one event API key. Do not configure two different server addresses for this setup. Each division has a separate Ref OS destination for its matches, scores, rankings and live field activity.

Configure both divisions

1. In the connector setup, choose Sync both TM divisions.
2. Confirm the shared TM server address and event key.
3. Choose Load TM divisions. If Ref OS has no division list, this setup can initialize it from TM. An existing list must contain the intended destinations.
4. In Division mapping 1, select the first TM division and its matching Ref OS division.
5. In Division mapping 2, select the second TM division and its different Ref OS division.
6. Check the names even if automatic matching selected them. TM and Ref OS numeric division IDs do not need to match.
7. Choose Review division 1 and Review division 2. Select a TM session for each when asked, then review again.
8. Choose Start both divisions.

What happens next

Snapshots run sequentially, one division after the other. Each follows its own 30-second qualification or 15-second elimination interval. A problem reading one division does not stop the other from retrying and syncing. Skills is imported once from the event-wide feed. Do not map both TM divisions to the same Ref OS division.

Switch the displayed division

Use the Division dropdown below the event header. Switching the displayed division, including on Matches, keeps the connector running. Check the division name before entering a violation or reviewing rankings. Q1 in one division is separate from Q1 in the other. Unassigned schedule represents older records without a division; it does not combine the division schedules.

Follow live matches

Current matches are green and upcoming matches amber. Open the Live fields dropdown for field activity. Use Jump to current match when you want to move; the app does not automatically take you away from a previous match. If connected mid-match, identity may remain unknown until TM sends an assignment event. Never choose a violation match solely because it is highlighted.

Elimination games

Quarterfinal, semifinal and final games can sync from TM, including distinct replay games grouped under the same pairing. Review the exact game before recording a ruling. A scored 0-0 result alone cannot tell Ref OS whether TM means a tie or a double DQ; use the explicit outcome control when confirmation is needed.

5. Log and review violations

Select the correct context

1. Confirm the event, League session and division.
2. Open Matches and select the match where the incident happened. You may stay on the previous match while the next one runs.
3. Open Log violation from the match details, or use the violation action from the appropriate team screen. The main match list may hide the floating logging button; open the match first.
4. Confirm your referee name in Logging as.

Complete the record

Select the team and verify its number. Select the correct match when the form asks for it. Choose the rule and the appropriate minor, major or inspection category. Enter a factual account: what happened, where and when, which team was involved, and the evidence supporting the decision. Attach an evidence photo if useful and permitted. Robot-on-file thumbnails help identify the robot; they are not a substitute for evidence of the incident.

Save and check

Choose Save violation. Confirm the entry appears in the correct match or team history. An eight-second Undo is offered after a new violation is saved. If you need a later correction, use the record's edit/delete actions where your role permits them. Do not log a second copy just to fix the first.

Review history

Use Teams to search for a team and open its history, or review violations in the match details. Repeated minor violations of the same rule can be surfaced for review. Ref OS is a recordkeeping aid: apply the current official manual, Q&A and event referee procedures rather than treating an automated count as a ruling.

Keep field events separate

Use Field Log for a timeout, field fault or other field incident. If flagging a replay, enter the reason. A field event, a replay flag and a team violation are different records; create only the records appropriate to what occurred.

Offline and pending saves

A new violation can remain in the device outbox when the connection is unavailable. Check connection health and pending items. Reconnect and confirm upload before assuming other devices can see it. Do not clear browser storage or sign out while important records are still pending. Ask an Admin to inspect failed sync items if retries do not complete.

6. Use the Robots tab

Purpose

Robots stores reference photos by team so inspectors and referees can identify robots and review changes. Photo completion does not certify an official inspection pass. The event Admin determines which views are required; inspect the labels shown in the current event.

Find a robot

1. Open Robots. Inspection users have a focused Robots and Rules workspace.
2. Search for the team number and open the team's robot record.
3. Verify the number before capturing or reviewing pictures. For a League, confirm the working session because sessions keep their own photos.

Capture the required views

Use the capture/upload actions on the team's photo slots. The usual sequence includes Front, Side, Back and Inspection Tag. A Lexan Diagram may be optional or required depending on the event configuration. Follow the required/optional labels in the app. Frame each view so the robot or tag is legible, review it, then save. Required views appear before optional ones in the camera sequence.

Completion and replacements

Use the photo counts and missing-view labels to identify incomplete records. To replace a view, retake or upload it in the same slot; the shared reference changes across devices. Check that you are replacing the intended view and team. Use any offered Undo promptly if a new upload was incorrect.

Sharing and offline access

Photos upload to the event's shared cloud storage. Images already viewed or downloaded on a device can remain available offline; an unseen image may be unavailable offline. New uploads can remain pending until internet access returns. Wait for successful upload before expecting another device to see the picture.

When a picture will not load

Read the failure message and use Retry. Confirm internet access and event sign-in. Missing-file, expired-link, permission and service errors need different follow-up; contact an Admin if retry fails. Do not delete the robot record to solve a loading problem.

7. Use Judging and interviews

Awards supported

The Judging workspace supports Sportsmanship and Energy nominations, finalists and candidate review. It does not replace the complete official judging process. Admins, Developers and Judge Advisors use the relevant judging controls; Emcees have viewing access without exports.

Create a nomination

1. Open Judging and select Sportsmanship or Energy.
2. Choose the nomination action and select the team.
3. Confirm the award, match/context and volunteer identity.
4. Select the observed criteria and record a specific example of the team's conduct. Add the supporting details requested by the form.
5. Save and confirm the nomination appears under the intended award and team.

Review candidates

Select the award and open a candidate's entry to review its nominations and associated information. Use finalist/shortlist controls where your role permits them. Admins can set candidate ranking order. A nomination count is supporting information, not an automatic award decision. Review violations in context and follow the event's judging procedures.

Interview Scheduler dropdown

Open Interview Scheduler - Experimental in Judging. Set the first time, a duration from 1 to 120 minutes, and the panel/location. Confirm the event timezone; unconfigured events use UTC. Select teams and add slots. Bulk slots are generated in numerical team order. Edit times, durations and panels before saving.

Schedule rules

Only one interview per team is allowed in the current schedule. Overlapping slots on the same panel are rejected. League sessions have separate schedules. Changing the timezone changes the display, not the saved interview instant. When another organizer has saved a newer version, reload the saved schedule and reconcile changes rather than overwriting their work. Unsaved edits may be discarded when reloading.

Export and handoff

Use the permitted nomination export controls to create the official Energy and Sportsmanship nomination PDF. Confirm names, teams and supporting text before sharing it with the Judge Advisor. Keep deliberation information within the authorized judging team. Judging changes require the event's cloud connection and appropriate permissions.

8. Event-day checks and troubleshooting

Before matches begin

Confirm the event and League session. Verify the roster, divisions and schedules. Check a team and match on two devices. Test role access, robot photo capture and a permitted sync check. In Event Command Center, use Pre Event System Test, Two Device Sync Test and offline readiness tools where available. Confirm the connector device can reach TM and the internet.

During the event

Monitor sync status, pending saves and the last successful update. Keep Android awake or Windows running in its tray. Verify both division mappings after setup. Keep referee entries on the match where the incident occurred. Use Request Help, volunteer status and the event contact directory for coordination.

TM cannot connect

Check the device is on a network that can reach the TM IP, confirm the port and Local TM API setting, and re-enter the event key. Check internet access and Ref OS administrator sign-in. For authentication errors, verify the device clock and key; developer credential setup is handled privately by the authorized developer.

Scores or rankings are stale

Confirm the event, session, division and sync status. Check whether TM itself has published the updated result locally. VEX Events publication may be delayed even when TM is current. Review both sync sources and their status before changing records. An empty source response should not erase existing data.

Only one division is available

Choose Sync both TM divisions, then Load TM divisions. Confirm TM actually has two divisions. Review both source and destination dropdowns. If using Android, install the latest rebuilt APK; an old app can retain the earlier interface. Windows uses the hosted website plus its native connector, so update the executable when native features change and refresh the website after deployment.

End of the event

Wait for pending violations and photos to upload. Export records and make a Backup all JSON from Event Command Center as needed. Stop both TM syncs or quit the Windows connector. Archive the event only when an authorized Admin is ready; archiving preserves records. Permanent deletion is a separate irreversible operation.

Keep credentials private

Do not share the TM event key with the general crew. Never include developer client IDs, client secrets, signing passwords or private keys in instructions, screenshots, exports or messages. This guide intentionally contains no credential values. Use the current official competition rules and the event's authorized procedures for rulings.